

CREATE A CLOUDFORMATION STACK FOR YOUR FLASK APPLICATION (EMPLOYEE DIRECTORY APPLICATION) INFRASTRUCTURE WITH THE MANAGEMENT CONSOLE

We are going to create the **infrastructure** for our **Flask Application** using the **management console**.

STEPS TO CREATE A STACK WITH THE MANAGEMENT CONSOLE

I) LOG IN TO YOUR **AWS ACCOUNT** AND SEARCH FOR **CLOUDFORMATION**.

II) IN THE **CLOUDFORMATION DASHBOARD**, CLICK ON **CREATE STACK**.

III) IN THE **CREATE STACK** PAGE, UNDER **PREREQUISITE – PREPARE TEMPLATE**, YOU CAN USE A TOOL CALLED **IAC GENERATOR** TO CREATE A **CLOUDFORMATION TEMPLATE** FROM YOUR EXISTING **INFRASTRUCTURE SETTINGS**. OPTIONAL: YOU CAN **CLICK ON IT TO FIND OUT**.

IV) UNDER **PREPARE TEMPLATE**, CLICK ON **CHOOSE EXISTING TEMPLATE**

V) UNDER **TEMPLATE RESOURCE**, SELECT **UPLOAD A TEMPLATE FILE**, UPLOAD YOUR TEMPLATE (USE THE TEMPLATE **6)_application_infrastructure_deployment_mgt.yml**), AND CLICK ON **NEXT**.

NOTE THAT WHEN YOU UPLOAD A TEMPLATE, IT IS AUTOMATICALLY STORED IN AN S3 BUCKET CREATED BY CLOUDFORMATION, AND YOU GENERATE AN S3 OBJECT URL FOR THAT TEMPLATE.

SUBSEQUENT TEMPLATES WILL ALSO BE STORED IN THAT S3 BUCKET.

VI) IN THE **SPECIFY STACK DETAILS** PAGE, UNDER **STACK NAME**, GIVE YOUR STACK ANY NAME. FOR ME, I WILL GIVE MY STACK THE NAME **employee-application-infrastructure**.

IN THE **PARAMETERS** SECTION, UNDER **AmiId**, LOCATE **DEFAULT**, REMOVE THE **DEFAULT AMI ID** THERE, AND PUT THE ONE FROM **YOUR OWN REGION IN YOUR ACCOUNT**.

TO GET THE **AMI** OF YOUR **OWN REGION**, OPEN **CLOUDSHELL** AND TYPE THE COMMANDS BELOW.

```
aws ec2 describe-images \
  --owners amazon \
  --filters "Name=name,Values=al2023-ami-2023*" \
           "Name=architecture,Values=x86_64" \
           "Name=virtualization-type,Values=hvm" \
  --query 'Images | sort_by(@, &CreationDate) | [-1].[ImageId,Name,CreationDate]' \
  --output table \
  --region eu-north-1 #change to your own region if you are not using eu-north-1
```

FILL IN YOUR **PARAMETER DETAILS** LIKE YOUR **DATABASE PASSWORD** (SHOULD BE SOMETHING LIKE **chinedu123**), YOUR **DATABASE SUBNET IDS** (CHOOSE TWO PRIVATE SUBNETS FOR HIGH AVAILABILITY), YOUR **EC2 SUBNET ID** (CHOOSE A PUBLIC SUBNET), AND YOUR **VPC ID**.

OBSERVE YOUR **TEMPLATE**, AND SEE THAT **EVERYTHING WE SPECIFIED IN THE PARAMETER SECTION IS WHAT IS STATED**. CLICK ON **NEXT**.

VII) IN THE **CONFIGURE STACK OPTIONS** PAGE, LOCATE **STACK FAILURE OPTIONS**, AND YOU CAN CHOOSE HOW YOU WANT **CLOUDFORMATION TO HANDLE YOUR INFRASTRUCTURE** IN TERMS OF **FAILURE**.

A) ROLL BACK ALL STACK RESOURCES: This means “If anything fails, undo everything.” **CloudFormation** deletes or reverses all resources created during the failed operation.

Suppose **CloudFormation** creates a **VPC** and an **EC2 instance**, both of which succeed, but the **Database fails**. **CloudFormation** then deletes the **VPC**, deletes the **EC2 instance**, and **stops everything**. This means that **nothing remains**.

B) PRESERVE SUCCESSFULLY PROVISIONED RESOURCES: This means “Keep the resources that succeeded, only roll back the failed parts.”

CloudFormation does not delete resources that were **created successfully**.

Suppose **CloudFormation successfully creates a VPC** and an **EC2 instance**, but the **Database fails**. **CloudFormation keeps the VPC** and **EC2 instance**, but rolls back only the failed database.

YOU CAN ALSO CHOOSE **YOUR OWN PREFERRED DELETION POLICY**.

VIII) LOCATE **CAPABILITIES**, CLICK ON THE **ACKNOWLEDGE BOX TO ALLOW CLOUDFORMATION TO CREATE IAM RESOURCES**. CLICK ON NEXT

IX) IN THE **REVIEW AND CREATE** PAGE, REVIEW YOUR CONFIGURATIONS AND CLICK ON **SUBMIT**.

X) IN THE **STACKS** PAGE, OBSERVE YOUR **INFRASTRUCTURE CREATION**.

IF THERE IS A FAILURE, **THEN A ROLLBACK WILL BE TRIGGERED**. ALSO **OBSERVE THAT ALL RESOURCES THAT WERE CREATED WERE ALL DELETED** AFTER WE HAD THE STATUS **ROLLBACK_COMPLETE**, WHICH TAKES ABOUT **10-15 MINUTES TO COMPLETE**.

IF THERE WAS NO FAILURE, THEN YOU SHOULD GET THE STATUS **CREATE_COMPLETE**.

UNDER THE **STACK INFO** TAB, OBSERVE THE STATUS OF YOUR **INFRASTRUCTURE BUILD**.

UNDER THE **EVENTS** TAB, OBSERVE THE STATUS OF **THE RESOURCES BEING BUILT**. A RESOURCE THAT IS SUCCESSFULLY BUILT WILL HAVE A **GREEN INDICATION** AT THE END OF THE GRAPH. IF THERE ARE ANY **ERRORS**, THEN A **ROLLBACK IS TRIGGERED**.

UNDER THE **RESOURCE TAB**, OBSERVE **THE RESOURCES** BEING CREATED AND THE **STATUS OF EACH RESOURCE** THAT IS CREATED.

UNDER THE **OUTPUTS** TAB, **OBSERVE THE NAMES** OF THE **RESOURCE** OUTPUTS YOU WISH TO SEE UNDER **KEY**; UNDER VALUES, THE NAMES AND **URLs** OF **SPECIFIC RESOURCES** YOU WISH TO SEE, AS WELL AS THEIR **DESCRIPTION** AND SO ON. YOU CAN COPY THE **IPV4 ADDRESS** OF YOUR EC2 INSTANCE, AND **PASTE IT IN ANOTHER TAB** TO SEE THAT **YOUR APPLICATION IS LIVE**.

XI) IF YOU WANT TO **TAKE A STEP FURTHER**, ADD A **NEW RESOURCE** IN THE **6)_application_infrastructure_deployment_mgt.yml** FILE. LOCATE CLOUDFORMATION, CLICK ON UPDATE STACK, CHOOSE TO USE YOUR EXISTING TEMPLATE, AND **CLOUDFORMATION** ADDS THE NEW RESOURCE TO YOUR **EXISTING INFRASTRUCTURE**.

